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Access to Data for Access to Justice: Unpacking Judicial Congestion Using an Event Log

(Authors' names blinded for peer review)

Problem Definition: Courts globally suffer from congestion, which undermines social welfare, economic growth, and access to justice. Despite the criticality of judicial operations, empirical research is scarce due to (a) complexity of court systems, characterized by constrained resources, increasing demand, prolonged waiting times, and cases lasting years, and (b) limited access to comprehensive data. This paper addresses these gaps by mapping the underlying judicial system and its operational dynamics as captured in court records.

Methodology: Using advanced natural language processing and machine learning techniques, we standardized 18 years of U.S. Federal District Court case dockets into an event log comprising millions of timestamped entries. The dataset presented in this paper lowers the barrier of access to data by capturing the temporal structure and multi-phase nature of judicial processes, as we demonstrate through preliminary descriptive analyses. The dataset thus facilitates investigations into judicial process modeling and workload patterns, helping to identify systemic bottlenecks and operational dynamics within the courts.

Managerial Implications: Our work establishes a robust, data-driven framework that provides descriptive insights into judicial operations. Moreover, it paves the way for further predictive, comparative, and prescriptive research. The dataset offers a foundation for exploring judicial process modeling, optimizing resource allocation, and assessing operational interventions, thereby opening new avenues for evidence-based strategies to alleviate court congestion and enhance access to justice.

Key words: Empirical Research, Non-Profit Management, Public Policy, Service Operations

1. Introduction

The court system is a unique operational environment, designed to produce justice. However, it suffers severe congestion worldwide, with negative implications on social welfare, economic development, and access to justice.

In the United States, for example, criminal case processing backlogs are estimated to cost an average of $9 million per jurisdiction per year in the U.S. (Schauffler and Ostrom 2020, Thomson Reuters 2021, p. 5). On the civil side, one study estimates that slow progress to trial costs businesses more than $10 billion over five years (American Arbitration Association 2017). Likewise, the backlog in U.S. immigration courts more than tripled between fiscal years 2017 and 2023, and asylum applicants fleeing violence and persecution wait in limbo for years until their status is adjudicated (U.S. Government Accountability Office 2023). The federal Speedy Trial Act, 18 U.S.C. § 3161, and state equivalents mandate specific timelines for resolving criminal cases but often fall short in practice, with systemic delays infringing upon constitutional rights (Cullen 2024).

Comparable challenges exist globally; for instance, Canadian courts often struggle to meet the "Jordan rule" (see Supreme Court of Canada 2016), a standard introduced by the Supreme Court of Canada to limit delays to 18 months for provincial courts and 30 months for superior courts. Despite these benchmarks, significant delays persist, leading to charge dismissals and eroding public confidence in the judicial system. These examples highlight the urgent need to address inefficiencies within judicial systems through evidence-based operational improvements.

The analysis of court operations is particularly important not only due to the courts' crucial role in society, but also their operational complexity. This complexity stems from multiple factors, including decentralization, high process variety, uncertainty about the duration and effort level required to resolve cases, and involvement of adversarial parties who may strategically delay the proceedings. The courts are also characterized by constrained resources, rising demand, long in-process waiting time, customers (litigants) who may depart the system in various stages, and an average Length of Stay (LOS) measured in years.

Though these unique features make the courts an interesting field for operation management research, work in the field still remains scarce, mainly due to the barrier of access to data. This issue is not exclusive to operations management research; empirical research in law also suffers from difficulties related to the availability, accessibility, and reliability of court systems data (Pereira et al. 2025, Voigt 2016). As observed in a recent report by the American Association for the Advancement of Science, "[C]ourts—unlike businesses—do not usually create data about their operations and analyze such data to improve the courts' practices" (see John and Nicholas 2022, p. 14). Moreover, while information about individual cases is accessible in the U.S., assembling data and records in bulk is cost-prohibitive at the federal courts' usual rate of $.10 per page (PACER). Access is even more difficult at the state level, as states and localities maintain their own separate record-keeping systems (Robinson and Gibson 2019). Similar challenges are observed globally, where limited institutional capacity and infrastructure often hinder the effective utilization of judicial data, as noted by Ramos-Maqueda and Chen (2025).

To facilitate research in this important field we present a unique dataset for operations researchers that gives a full granular operational picture of 18 years of civil and criminal litigation for the Northern District of Illinois (NDIL). This district is the third-largest and one of the busiest federal district courts in the United States, covering 10,100 square miles and 9.3 million people, including the city of Chicago (www.ilnd.uscourts.gov). In all, the dataset comprises 175,268 cases, with 4,811,483 timestamped activities, coupled with case-level static attributes, judge identification, and activity-level dynamic attributes. We share this data with the operations research community to provide an opportunity to drive social impact operations research (Jónasson et al. 2022) and to use it as a fertile ground for understanding highly congested court systems.

This work is based on a deep collaboration with the Systematic Content Analysis of Litigation Events Open Knowledge Network (SCALES OKN or SCALES) project, a multidisciplinary research group focused on increasing access to court records and analytics. The administration of NDIL granted the SCALES project a fee waiver for court record access. We thus assembled a comprehensive set of *docket sheets* from the court for all cases filed between 2002 and 2020. A docket sheet provides a contemporaneous chronological listing of dates and litigation events in each case, the names of the parties, lawyers, law firms, judges, and other case-level information. Using the raw docket sheet text, we developed Machine Learning (ML) models that applied standardized litigation event labels to the original text, which often varied across cases in describing the same litigation activity. We further refined the labeling scheme and aggregated and transformed the labeled data into event logs - a detailed record of the sequence of activities within a system, which serves as a foundational element in process analysis and process mining(van der Aalst 2016).

This paper contributes to advancing empirical research into judicial delays by addressing the challenges of working with unstructured data and transforming it into actionable insights to improve access to justice. We begin in Section 3 by describing the U.S. Federal District Court system and adjudication process, emphasizing their complexity and unique characteristics. Building on this foundation, in Section 4.1 we conceptualize the judicial process by segmenting it into sub-processes. We further develop a structured approach to transform raw docket data into a standardized event log. This transformation involves employing ML classification models for NLP and designing a nesting scheme that maps labels to activities and attributes, enabling a standardized view of the judicial process. In Section 5 we present a validated event log of court operations, spanning 18 years of data from the NDIL Federal District Court, offering a unique resource for examining court operations at a granular level. This is the first such log on U.S. courts, and the most extensive validated court operations event log dataset publicly available. In Section 6 we outline specific research directions that leverage the new data set to lay the foundation for future studies on congestion alleviation, resource optimization, and policy interventions within complex service systems like the courts. We close with some concluding remarks in Section 7.

2. Literature Review

Court operations as a field of study have gained increasing attention in the OR/MS community, to address critical challenges such as resource allocation, scheduling, performance measurement, and process analysis. For example, Dimitrova-Grajzl et al. (2012) analyze Slovenian courts based on a panel of 396 court-year observations, and find that increasing the number of serving judges does not necessarily drive court output because incumbent judges may reduce their productivity, and court output is primarily driven by the demand for court services. Mukherjee and Whalen (2018) suggest a priority queuing framework to model the distribution of time to disposition in the supreme courts of the United States, Canada, and Massachusetts. Their empirical analysis, using case-level data, reveals universal queuing patterns, where the vast majority of cases are resolved at a very early stage, leaving a small share of outlier cases that take an extremely long time to resolve. Falavigna et al. (2015) and Peyrache and Zago (2024) employ methods such as Data Envelopment Analysis and production frontier simulations to derive efficiency measures and the impact of policy changes on the Italian judiciary from court-year observations.

While the papers mentioned above yield important aggregate insights, using case-level and court-year data implies that they lack the internal dynamics of case progression in their analysis. Further reinforcing the limitations of aggregated data analysis, Mazzocchi et al. (2024) conduct an efficiency analysis of the Italian judicial system using a two-step analytical framework. Their findings indicate that judicial efficiency is not significantly impacted by the procedural steps of case progression. Gupta and Bolia (2023) develop an integer programming model to optimize judge assignments in Indian district courts, demonstrating that judicial reallocation can increase case disposals and improve clearance rates without additional resources. Their findings support staffing-based interventions as a means of reducing backlog. However, like other studies in this category, their approach relies on aggregated clearance rates and case disposals, rather than tracking detailed procedural movements within cases. Azaria and Shamir (2024) use two instrumental variables to show what they term as 'congestion vortex', where delay increases the number of decisions in motions, adding to the judge's workload. However, their data focuses on hearings and decisions, leaving out meaningful events in the case life cycle, as captured by the data set described in this paper.

Complementing these works, recent studies have attempted to bridge this gap using event-level data and process mining techniques. Unger et al. (2021) focused on the São Paulo Court of Justice, mapping actual case flows and uncovering that only 3% of cases follow the most common workflow, highlighting the unstructured nature of legal proceedings. Martino et al. (2021) extended this approach by integrating graph analysis with process mining to detect temporal outliers in online civil trials, pinpointing procedural steps where significant delays occur. Vercosa et al. (2023) applied a hybrid approach—combining process mining with machine learning—to predict lawsuit duration in Brazilian labor courts by clustering procedural event sequences. Pernici et al. (2024) further contribute by directly extracting and analyzing event logs from a court's digital register to assess the timeliness of civil trials. While these studies use event logs to map judicial workflows, they typically rely on datasets that are limited in size or scope. Moreover, these logs tend to overlook critical information like lawyer filings, judge reassignment events, and case-level complexity drivers.

A second gap in the literature arises from the focus on judicial environments that are inherently less complex than U.S. federal district courts. Engel and Weinshall (2020) examine small claims in Israeli Magistrate Courts, demonstrating that reduced caseloads lead to improved outcomes, in terms of settlement rate and reduced biases (Kricheli-Katz and Weinshall 2023); yet, these simpler procedures do not capture the unstructured, heterogeneous processes encountered in US federal courts. Coviello et al. (2015) track Italian labor dispute cases to reveal the negative impact of task juggling on case completion, as judges require time to reacquaint themselves with the details of cases in high workload environments. Bray et al. (2016) describe in their paper a data-driven intervention in the Italian Labor Court of Appeals, modeling case prioritization using multi-armed-bandit to propose different scheduling schemes based on case completion hazard rates. However, appellate courts experience significantly lower levels of complexity and uncertainty in their proceedings compared to courts of first instance or general jurisdiction.

Along the line of these studies, Azaria et al. (2023, 2024) demonstrate how OM-based lead time reduction case processing times in Israel's Jerusalem District Court, as part of a pilot implementation. While the results show a significant reduction in the trial phase LOS for civil cases, their empirical analysis focuses on the impacted cases - civil cases with trials - without capturing any potential adverse effects on other cases in the court.

In more recent studies, Bakshi et al. (2024) hypothesize that judicial congestion can be analyzed as a case-management queue, and show that congestion can be mitigated by optimizing scheduling guidelines. They examine congestion in the Supreme Court of India as a final appeals court, handling high caseloads with a two-stage process. They employ data-driven queuing simulation, calibrated with empirical data, to model court congestion and evaluate operational interventions. The model accounts for two-stage queues, shared judicial capacity, and external delays. Similarly, Freund and Weng (2024) use administrative reports from U.S. Immigration Courts to calibrate a stylized queuing model to demonstrate that a dedicated docket designed to fast-track immigration cases improves efficiency compared to a mixed cases docket. These courts environments also lack the procedural complexity characteristic of U.S. federal district courts.

Clarke and Metzler (2024) use data on the duration of criminal cases over two decades across Canadian provinces to analyze the performance of the criminal justice system using queuing theory. They find that court congestion has explanatory power for the rising population in pretrial custody. Ata et al. (2025) also examine congestion and pretrial detention, this time in Cook County Jail and the Illinois Department of Correction. They find that reducing unnecessary court visits, funding low-level detainees' bonds, implementing split sentencing, and addressing detainees' perceived costs of prison, could significantly lower pretrial incarceration, cut costs, and ease court congestion. Adding another dimension, Li et al. (2024) present an innovative approach for incarceration-diversion program management by combining machine learning with queuing theory. Their decision support system integrates process-level data to predict program census and staffing needs while addressing challenges such as right-censoring in length-of-stay predictions. These works, focused on other parts of the adjudication system and their relationship with court congestion, reinforce the broader methodological insight that detailed, dynamic data are essential for understanding complex legal processes.

Taken together, these studies reveal two critical gaps. First, while these methods compute overall performance indicators from aggregated data, they do not capture the detailed, sequential procedural movements and high variance between individual cases. Second, these investigations mostly focused on judicial or diversion settings with relatively simplified procedural frameworks. Addressing these shortcomings, the present study leverages a uniquely granular dataset that captures every procedural movement, precise timestamps, and rich case-level metadata within U.S. federal district courts, offering a more nuanced understanding of judicial bottlenecks and operational dynamics in a complex legal environment.

3. Institutional Background

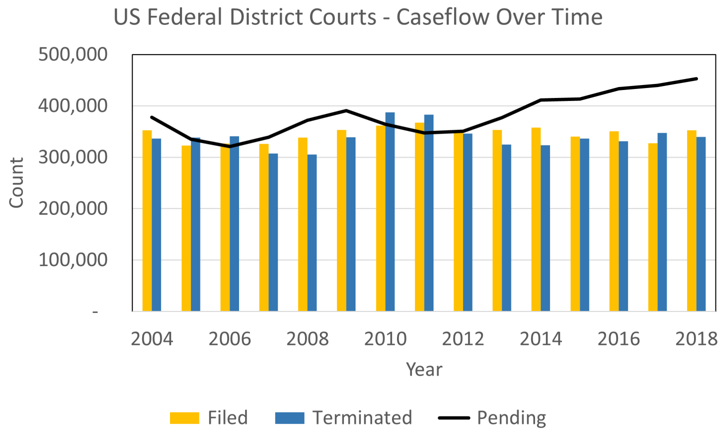
3.1. The U.S. Federal District Courts

The United States operates under a dual court system comprising federal and state courts, each with distinct structures, jurisdictions, and functions. There are three levels of federal courts: district courts (the trial courts), circuit courts (the appeals courts), and the Supreme Court of the United States (the appellate court of last resort). There are 94 district and 13 circuit courts, with geographical boundaries set by federal statute. Each state has at least one district court; each district court belongs to one circuit court. District courts are courts of general jurisdiction and handle both criminal and civil cases. In order to be filed in the federal court, as opposed to the state trial court in the same geographic area, the case must meet a set of criteria. Criminal cases must allege violations of federal criminal statutes. Similarly, civil cases must include federal statutory or Constitutional claims. Civil cases involving only state law claims can be brought in federal court if all parties are domiciled in different states and the amount in controversy is greater than $75,000, or if the United States is a party to the case (www.justice.gov).

Judges in all federal courts are nominated by the U.S. President and confirmed by the U.S. Senate. They hold lifetime appointments and may be removed only via impeachment proceedings in Congress. There are currently just over 1,000 federal district court judges, also known as Article III judges, as well as almost 600 federal magistrates, who are employees of the courts and are designated to handle certain civil and criminal matters, depending on the case type and local practices of the court (www.abalegalprofile.com; crsreports.congress.gov).

When a new case is filed in court, it is assigned to a district judge, usually at random, though details of the case assignment procedure, such as the extent to which current load of the judge is

Figure 1 **Federal District Courts annual count of all cases filed over time.**



taken into consideration, vary by jurisdiction (www.uscourts.gov). Some or all parts of the case may also be handled by a magistrate judge. In criminal cases, magistrates typically make bond decisions and try and sentence misdemeanor defendants. In civil cases, magistrates can resolve discovery disputes, conduct mediation, issue a recommendation to the district judge on a dispositive motion, and even sit as the sole judge for the entirety of a case with the parties' consent. While the division of labor between district and magistrate judges varies according to case types, party preferences, and local rules and practices, in general terms, magistrates handle simpler, lower-stakes matters as compared to their district counterparts (www.proquest.com).

In the 12 months ending on September 30, 2023 (the most recent record-keeping period at the time of writing), the 94 federal district courts handled approximately 1.5 million cases. Of those, 405,878 were new case filings, up 18 percent from the previous 12 months. These new filings joined 759,069 cases that were previously filed and remained open throughout the year, while 366,048 previously filed cases that closed during those 12 months. Across all cases, about 83% were civil and the remainder were criminal (www.uscourts.gov.)

Figure 1 shows the annual rate of arrival of cases to the federal district courts in the years 2004-2018. The data include all civil and criminal cases, collected from the Administrative Office of the U.S. Courts statistical reports, calculated for the time period ending September 30th annually (www.uscourts.gov.) There are evident fluctuations in filed cases over time, possibly related to macroeconomic factors (e.g., output, consumption and inflation; Bachmeier et al. 2004). What is also evident is that while the termination rate seemingly followed the filing trend, it was fairly consistently behind it, causing an increase in pending cases.

In terms of measures of performance, the Civil Justice Reform Act of 1990 requires twice-annual reporting of the rate at which judges dispose of civil cases and motions. In addition, for all case types, the Administrative Office of the U.S. Courts tracks the number of cases filed, pending, and terminated each year and makes statistical tables publicly available (Administrative Office of the U.S. Courts 2024). Though these reporting mechanisms are not designed to function as performance evaluations, they roughly track the performance metrics commonly used by courts around the world, including case clearance rate, time to disposition, and case backlog (Dakolias 1999, International Consortium for Court Excellence 2020). Notably, the main indicator of effectiveness of a court used in the literature is time to disposition (Marciano et al. 2019), which we refer to in this paper as LOS.

3.2. The Judicial Process

Figure 2 provides a highly simplified illustration of major steps in the litigation process in civil and criminal cases handled by federal district courts. Civil cases typically begin with the plaintiff's filing a complaint, which provides a short description of the dispute and the legal claims. Defendants may then file an answer to the plaintiff's claims or move to dismiss the case. After an answer is filed if the case is not dismissed, the case moves to the discovery phase, when both parties are required to exchange information and evidence that is relevant to the claims and the defense. Discovery generally takes place outside of the court, but the judge may be called upon to resolve disputes about the scope and relevance of discovery requests. After discovery, either party may file the motion for summary judgment, in which the party presents their full case and requests a judgment in their favor before the trial. If this motion is not filed or if it is not granted, the case proceeds to trial, where a verdict is rendered for one of the parties. Notably, a settlement between the parties can happen at any time during the case life cycle, regardless of its stage; the vast majority of civil

Figure 2 Civil and criminal process map (highly simplified).

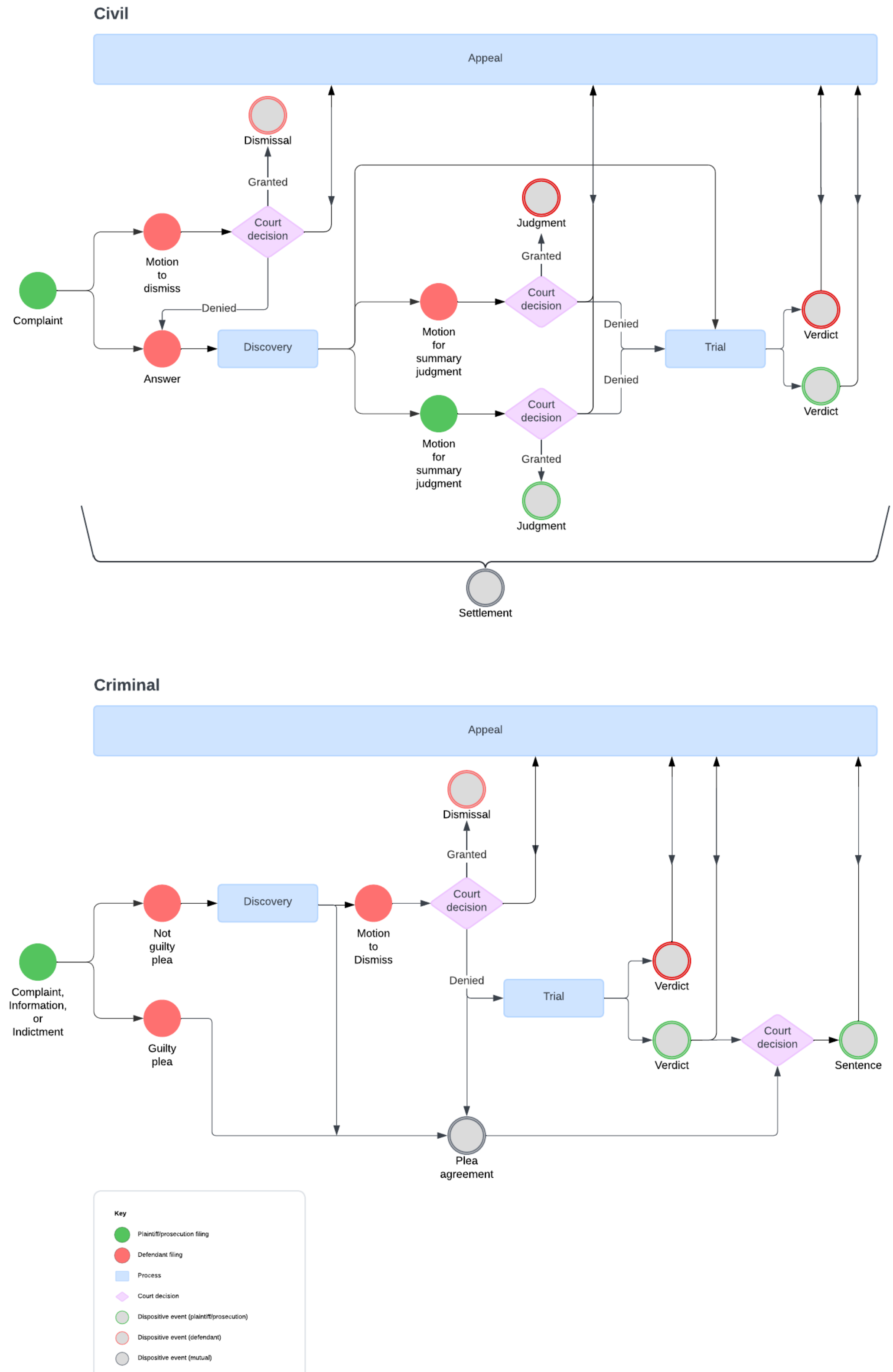
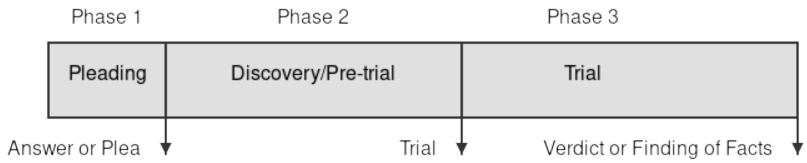


Figure 3 The three major phases of the judicial process.



cases are settled prior to the verdict. Moreover, litigation may be paused at various junctures for a party to appeal a decision made by the district court judge to the appellate court.

Criminal cases generally begin with the filing of a complaint, information, or indictment, which outlines the charges against the defendant. The defendant may enter a plea of either guilty or not guilty. A not guilty plea moves the case forward into the discovery phase, during which both the prosecution and defense exchange information and evidence relevant to the charges. During or after discovery, the defense may file a motion to dismiss, requesting the court to terminate the case due to insufficient evidence or legal defects. If this motion is denied, the case proceeds to trial, where a verdict is reached. If the defendant is found guilty, the court issues a sentence based on the applicable laws and guidelines. However, as with civil cases, criminal cases are most commonly resolved outside of trial. A plea agreement, where the defendant agrees to plead guilty to certain charges in exchange for concessions from the prosecution, is a frequent resolution that avoids the need for trial. Similar to civil cases, the parties may negotiate a plea agreement at many stages in the case, and criminal cases may also be paused for appeals at various stages of the process.

It is useful to think of the process for both civil and criminal cases, as having three major phases illustrated on Figure 3. The first phase is *pleading*. It starts on the date when the case is filed, usually with a complaint in civil cases and a complaint, information, or indictment in criminal cases. This phase ends with an answer from the defendant in civil cases or a plea in criminal ones. The second phase, the *discovery* or *pre-trial* phase, starts with the answer or plea and ends when the case proceeds to trial. As noted above, discovery occurs primarily outside of the court. Nevertheless, this phase can involve many activities that require the judge's involvement, including resolving discovery disputes and ruling on post-discovery dispositive motions that seek to terminate the case prior to trial. The third phase is the *trial* phase. It starts when the case proceeds to trial and ends with a "disposition" (verdict in civil cases/ sentence in criminal ones).

In practice, most cases are resolved during the initial pleading phase, making this a key characteristic of the process dynamics. These cases are either settled, dismissed, receive a default judgment, or are transferred to other jurisdictions. As Alexander et al. (2024) found in their study of civil cases, most cases (about 60 percent) that exit litigation at this early stage involve very little formal adjudicatory activity by judges. However, in phase 1 as in later phases, judges may exert influence in other, less formal ways, including by requiring parties to report on their progress toward settlement, setting an aggressive calendar, and otherwise "nudging" the case toward resolution. For example, as U.S. Supreme Court Chief Justice Roberts (2015) has observed, "a well-timed scowl from a trial judge . . . can go a long way in moving things along crisply."

As is the case with phase 1, most cases that enter phase 2, the discovery or the pre-trial phase, are finalized during this phase. Furthermore, even some cases that make it to the third phase, the trial, do not necessarily proceed to a disposition, but are instead resolved through a settlement by the parties. All in all, only a very small fraction of all cases reach this final stage of litigation and end with a verdict (Chang and Klerman 2022). This phenomenon is termed by legal scholars the "vanishing trial" (Galanter 2004). This general phenomenon certainly holds for our NDIL dataset - as discussed in Section 5 and shown in Table 5, only 47.2% of all cases in our dataset enter phase 2, just 1.8% enter phase 3, with only 1.1% completing all three phases.

Another unique characteristic of the litigation process is that cases may narrow over time, as certain parties or claims can be either resolved or dismissed, while others proceed to later litigation stages. Moreover, not only can cases (or their parts) be resolved or terminated at any point during the case life cycle, but they can also be subject to different drivers – party resolutions, court decisions, and combinations of both.

These unique features the litigation process make it quite complex to model and to analyze. Moreover, the complexity of the process and the extremely long LOS, make case-level analyses not only less informative but also prone to biases. A more useful analysis requires much more granular view of various activities comprising a case. For this reason, in section 4 we discuss the construction of a granular event log for each case from federal court docket sheets. Event logs are particularly useful in environments where processes are flexible, heterogeneous, or "spaghetti-like," as they accommodate variations and unexpected behavior (D'Castro et al. 2018). This flexibility is essential for systems like courts, where human decisions and procedural complexities often defy standardized modeling (Unger et al. 2021).

4. Creating an Event Log of a Court

Creating litigation event log is the main objective of this paper. The event log for NDIL is available on the SCALES-OKN website (scales-okn.org).

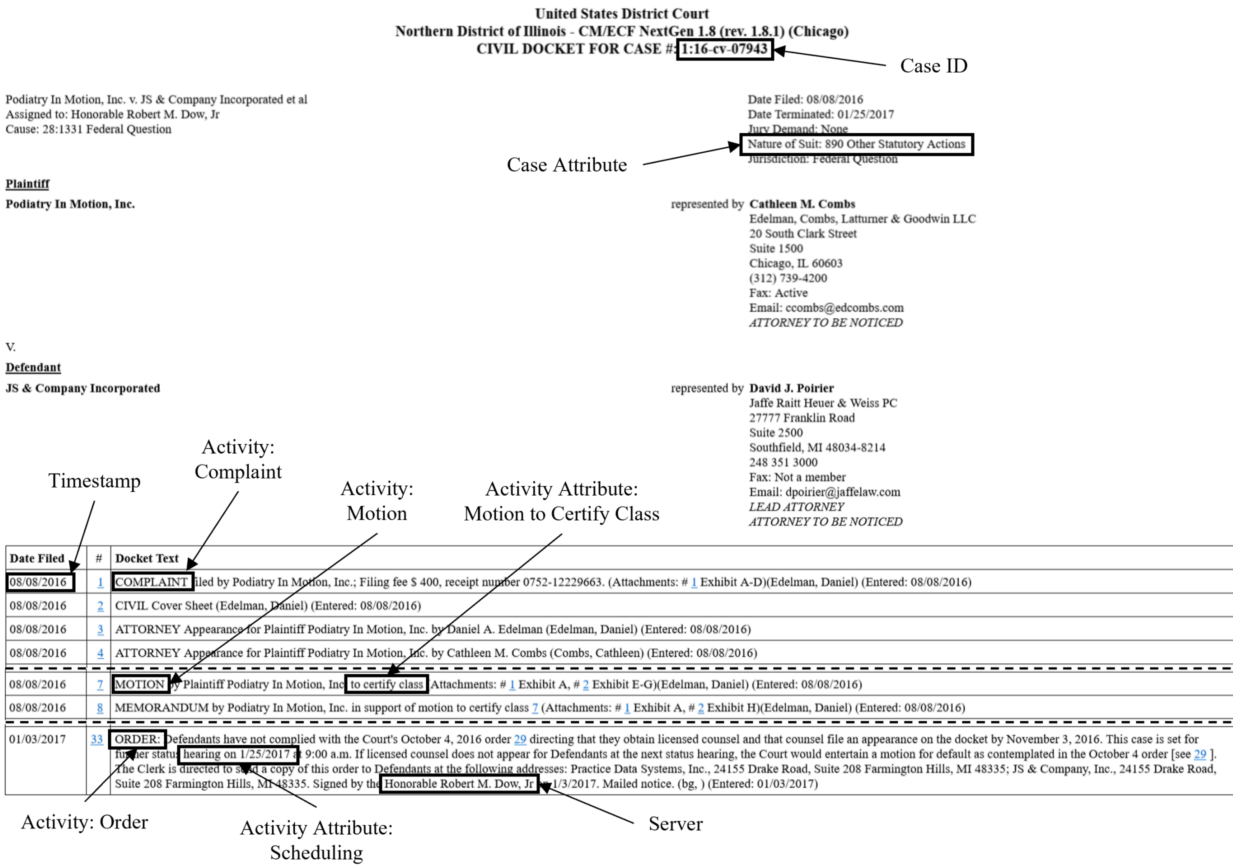
The event log data structure enables viewing the court system via an operational lens by capturing the judicial resources and activities involved in the progress of cases. Methods such as statistical analysis and process mining allow descriptive analytics of the data, including the depiction of case flow through the system and measuring its operational performance (van der Aalst 2016). Machine learning methods can be used for predictive analytics, including forecasting flow times and identifying cases that may be subject to long delays. Methods combining process mining with data-driven simulation, see, e.g., SiMLQ, allow comparative analytics by creating, calibrating, and validating a process simulation while maintaining queuing causal model (Chapela-Campa et al. 2025).

In subsection 4.1 we explain an event log as a list of events (case IDs, activities, and time stamps). In subsection 4.2 we detail how we aggregated the individual cases' docket sheets to a comprehensive dataset including all cases with standardized terminology. In this stage, we started with the docket sheets, extracted the relevant text, and developed ML models to apply standardized labels for each element. Finally, in subsection 4.3, we describe how we transformed this comprehensive dataset to create the event log for NDIL.

4.1. What is an Event log?

An event log is a detailed record of (a) *Entities*, here Cases, that are captured via a unique identifier, *Case ID*. These entities go through (b) *Activities* occurring within the process. The activities'

Figure 4 Docket snippet from case #: 1:16-cv-07943.



United States District Court
Northern District of Illinois - CM/ECF NextGen 1.8 (rev. 1.8.1) (Chicago)
CIVIL DOCKET FOR CASE #: 1:16-cv-07943

Podiatry In Motion, Inc. v. JS & Company Incorporated et al
Assigned to: Honorable Robert M. Dow, Jr
Cause: 28:1331 Federal Question

Date Filed: 08/08/2016
Date Terminated: 01/25/2017
Jury Demand: None
Nature of Suit: 890 Other Statutory Actions
Jurisdiction: Federal Question

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LEAD ATTORNEY
ATTORNEY TO BE NOTICED

Date Filed	#	Docket Text
08/08/2016	1	COMPLAINT filed by Podiatry In Motion, Inc.; Filing fee $ 400, receipt number 0752-12229663. (Attachments: # 1 Exhibit A-D)(Edelman, Daniel) (Entered: 08/08/2016)
08/08/2016	2	CIVIL Cover Sheet (Edelman, Daniel) (Entered: 08/08/2016)
08/08/2016	3	ATTORNEY Appearance for Plaintiff Podiatry In Motion, Inc. by Daniel A. Edelman (Edelman, Daniel) (Entered: 08/08/2016)
08/08/2016	4	ATTORNEY Appearance for Plaintiff Podiatry In Motion, Inc. by Cathleen M. Combs (Combs, Cathleen) (Entered: 08/08/2016)
08/08/2016	7	MOTION by Plaintiff Podiatry In Motion, Inc to certify class Attachments: # 1 Exhibit A, # 2 Exhibit E-G)(Edelman, Daniel) (Entered: 08/08/2016)
08/08/2016	8	MEMORANDUM by Podiatry In Motion, Inc. in support of motion to certify class 7 (Attachments: # 1 Exhibit A, # 2 Exhibit H)(Edelman, Daniel) (Entered: 08/08/2016)
01/03/2017	33	ORDER: Defendants have not complied with the Court's October 4, 2016 order 29 directing that they obtain licensed counsel and that counsel file an appearance on the docket by November 3, 2016. This case is set for further status hearing on 1/25/2017 at 9:00 a.m. If licensed counsel does not appear for Defendants at the next status hearing, the Court would entertain a motion for default as contemplated in the October 4 order [see 29]. The Clerk is directed to send a copy of this order to Defendants at the following addresses: Practice Data Systems, Inc., 24155 Drake Road, Suite 208 Farmington Hills, MI 48335; JS & Company, Inc., 24155 Drake Road, Suite 208 Farmington Hills, MI 48335. Signed by the Honorable Robert M. Dow, Jr on 1/3/2017. Mailed notice. (bg,) (Entered: 01/03/2017)

names describe the relevant actions. Examples of court system activities include complaint and verdict (see Figure 2). And (c) the *Timestamp* (event) at which a specific entity interacted with a specific activity. In the court context, timestamps are recorded as the date when an activity occurred, as recorded in the court docket.

In addition to the case ID, activities, and timestamps, event logs often include metadata about the entities and activities involved. The event log we provide includes the following:

- *Server ID*: The judge (or judges) responsible for the event (standardized judge ID code).
- *Case attributes*: Case metadata and attributes (e.g., nature of suit, number of plaintiffs).
- *Activity attributes*: Information unique to an activity that distinguishes it from other activities of the same type (e.g., unopposed, dispositive).

As Figure 4 shows, the source for these elements is each case's docket sheet, which is the chronological record of litigation activity made in real time as the case progresses. Docket sheets are available for download in HTML format from the federal courts' Public Access to Court Electronic Records (PACER) information system.

We describe the construction process of the event log dataset in two stages. In the first stage, see subsection 4.2, we extracted the relevant text from the dockets sheets and developed ML models to apply standardized labels and entity disambiguation for each element. In the second stage, see subsection 4.3, we transformed the raw data generated in the previous stage to an event log. This means, among other things, creating *Case attributes* based on the scraped metadata and designing the scheme to translate the labels to *Activities* and *Activity attributes*, which allows a standardized description of the events ontology.

4.2. From Docket Sheets to Datasets

As substantial parts of docket sheets are maintained as free text and have many variations among different courts, we must first transform them to structured data with standardized components. We created three tables with structured data: **Cases Metadata**, **Standardized Judges Ontology**, and **Labels Ontology**. All data are available on the SCALES-OKN website (scales-okn.org).

As shown in Figure 4, docket sheets have two main sections: header fields at the top that record the judges, parties, lawyers, and claims or charges in the case, and a chronological index below that lists dates on which activities took place along with a free text activity description. The case attributes stored in the Cases Metadata table are drawn from the header fields. The judges in the Standardized Judges Ontology come from the header fields as well as additional judges' names mentioned in the index of docket entries. The Labels Ontology table is drawn from the docket entry text. We discuss each below.

Parsing the header fields to assemble the first table, **Cases Metadata**, is relatively simple, as the header information tends to be fairly well standardized across judges, cases, and courts. For example, the case attribute "Nature of Suit," shown in Figure 4, captures the main type of claim in a civil case. Plaintiffs must choose one option from a fixed list when filing their complaint.

The second table, **Standardized Judges Ontology**, presents more of a challenge. While the header fields record the names of a district and magistrate judge in each case, those fields are updated each time a judge changes. Therefore, those fields only capture the most recent judge(s) involved in the case at the time the docket sheet is accessed. To capture the full picture of judicial involvement, we build a Named Entity Recognition (NER) workflow to mine the text of the docket entries and extract, disambiguate, and standardize the names of all judicial officers with a role in a case over its entire life cycle (Pah et al. 2021).

Our approach used the spaCy non-transformer architecture (Honnibal et al. 2020) to train a model on approximately 1.4 million docket entries containing approximately 3.4 million annotated judge names, which we identified by the presence of titles such as "Judge," "Magistrate," and "the Honorable." After using the model to extract all probable judge names and validating model results, we used similarity measures (fuzzy matching based on the Levenshtein distance) to determine whether different name variations should be collapsed into single judge identifiers. This similarity allowed, e.g., "Matthew Kennelly," "Matthew F. Kennelly", and "M. F. Kennelly" to all represent a single judge entity, represented by a unique standardized judge identifier (SJID), *Server ID*. We further used the official list of Article III judges from the Federal Judicial Center (Federal Judicial Center 2024) to label judges according to their status, e.g. District (Article III) versus Magistrate.

Constructing the third table, **Labels Ontology**, was even more complex. A brief detour into the data generation process behind the docket entry text provides useful context. Though all federal district court docket sheets are available for download via the centralized PACER portal, they are actually generated via 94 separate local electronic docketing and case management systems, one for each district court. Parties or their lawyers, as well as court clerks and judges, use those 94 separate local systems to file and label documents and record case activity.

This distributed structure produces substantial variation across cases, judges, and courts. The U.S. District Courts for the Northern and Southern Districts of Georgia, for example, offer plaintiffs in civil cases seven and five different menu options, respectively, for labeling a "Case Initiating Document" (source: Northern District www.gand.uscourts.gov; Southern District www.gasd.uscourts.gov). There is no substantive difference between the types of documents that may be filed in the two districts; this variation stems from different organizational and record-keeping choices at the local court level.

In addition to choosing from a menu of labels, parties, lawyers, clerks, and judges often enter free-text activity descriptions. For instance, docket entries labeled "Notice" in two different cases might say in the free-text description: "The plaintiff and defendant have agreed to settle" and "The parties reached a mutual resolution of all claims." Both refer to settlement, the same underlying litigation activity, though the activity descriptions are completely unstandardized.

Thus, to move from docket to data and create the **Labels Ontology**, the docket entry information needed to be standardized. Working with a team of legal subject matter experts and computer scientists, we built a set of ML classifiers to extract and label litigation events, which we later transformed into *Activities* and *Activity attributes*, along with *Timestamps* (see subsection 4.3).

To generate the event labels, we first developed a litigation ontology to capture all substantive activities documented in the docket entries, such as the parties' filing of a motion or a judge's ruling. In addition, we added labels indicating the entry role in the process, such as case opening, or its specific circumstances such as 'bilateral' or agreed-upon by the parties. Lastly, we created process-oriented labels, such as 'deadline extension' and 'rescheduling' notices.

When creating these labels, our guiding principle was that for a docket entry to be considered a label, it needs to impact flow, either via routing or timing. The ontology thus captures the full range of substantive, procedural, and administrative activities in each case and reflects the progress of cases through the system.

To operationalize the ontology, we built a set of ML models, collectively referred to as the *ML classifier*, that took as their input the raw docket entry text and applied the relevant set of labels. We used Microsoft's open source "large DeBERTaV3" model, pre-trained on general English language usage, and further trained the model on 11 million docket entries to expose it to legal terminology. For each label, we created a training set of between 1,000 and 4,000 hand-annotated docket entries. We annotated more entries for labels that had more textual variation in the underlying docket entries. The 'settlement' label, for example, required a larger set of hand-annotated entries to capture the substantial variation in docket entry text, whereas the 'motion' and 'complaint' labels required fewer, since the underlying text descriptions were more consistent. After validation and testing, we deployed the classification models to generate a set of standardized labels applicable to docket entries such as those shown in Figure 4. This is the main output of the classification process - a set of labels, created to map all the relevant substantive, procedural, and administrative steps of the process, identified via ML classification for each docket entry.

4.3. From Datasets to Event Log

To move from datasets to the event log, we began with the three tables produced during the initial stage described in subsection 4.2: **Cases Metadata**, **Standardized Judges Ontology**, and **Labels Ontology**. From these we constructed our event log dataset.

The Case Metadata dataset serves as the primary record for case-level information collected from each case's docket sheet header in the PACER system. It includes highly detailed information, from the Nature of Suit Code to the contact details of each party and counsel involved in the case. This dataset enabled the creation of the granular *Case attributes*. When constructing *Case attributes*, some of these are drawn as-is, such as 'case type' (civil or criminal) and 'city' (Chicago or Rockford, the two divisions courthouses), but others are engineered features. The decision of what information to include and which features to construct was guided by the role that *Case attributes* play in the event log. These static features (i.e., their values are invariant for all events linked to a given case) are meant to indicate the complexity of a case, such as the number of parties involved in the case, or the share of which who are individuals rather than organizations (see complete feature list and definitions in Appendix C).

The bulk of the work of creating the event log involved reshaping the Labels Ontology table. The table began as a flat label structure, which presented all labels on a single level without any hierarchical organization. However, in unstructured processes, activities often exhibit high variability due to contextual differences or noise, resulting in overly detailed models. Based on

Table 1 Structure of the event log dataset.

Case ID	Timestamp	Activities	Server ID	Activity Attributes	Case Attributes
1:02-cv-00430	2002-02-01	complaint	SJ000169	opening	Employment
1:02-cv-00430	2002-02-19	summons	SJ000169		Employment
1:02-cv-00430	2002-03-08	motion	SJ000169	motion to dismiss	Employment
1:02-cv-00430	2002-12-05	motion	SJ000169	time extension	Employment
1:02-cv-00430	2003-01-16	minute entry	SJ000169	unopposed	Employment
1:02-cv-00430	2003-01-16	order	SJ000169	rescheduling	Employment
1:02-cv-00430	2003-08-22	settlement	SJ000169	dispositive	Employment

the principles presented by de Murillas E (2019), the construction of the set of activities should be done so that it keeps the tension between representing meaningful traces and being able to interpret them. Our challenge was therefore to minimize the number of events per case, but still retain sufficient detail to capture any notable patterns to achieve what de Murillas defines as "interestingness." We achieved this by creating a hierarchical nested structure, with a limited set of broader activity types, with associated attributes that maintained access to specific details (e.g. 'motion' and 'motion to certify class', see Figure 4). This reduced redundancy, improved clarity, and struck a balance between representation and interpretation.

To that end, from the full set of labels, we developed a nesting scheme separating 26 *Activities* with an associated set of 60 *Activity attributes* (see complete list and definitions in Appendix B). *Activity attributes* are additional information that distinguish one event from another, impacting routing or timing. This means that *Activity attributes* are either subtypes of *Activities* or they indicate a feature related to the flow. For example, the activity 'order' could have attributes such as 'default judgment,' indicating the type of order, and 'dispositive,' indicating that this order terminates a claim, a party, or a case. Nesting activity types preserves the descriptive power of event logs while mitigating the challenges posed by excessive detail. It allows analysts to focus on overarching process dynamics without losing the richness of unstructured data.

After separating *Activities* from *Activity attributes*, we still have to address two issues when restructuring labels into events. First, some docket entries describe multiple events. For example, one entry could describe the first day of a trial taking place but also the settlement of the parties' claims, meaning that the rest of the trial was canceled. We decided to treat these two *Activities* (trial and settlement) as separate events that occurred at the same time. Second, after implementing the nesting logic, we ended up with entries that were labeled only as *Activity attributes* and not as *Activity*, i.e., "orphaned" activity attributes, as a result of ML classifier errors. We were able to match half of these orphaned activity attributes and merge them into same-day *Activities*. For many other orphaned attributes, we were able to impute the *Activity* after careful consideration and manual review. The remaining orphaned *Activity attributes* were filtered out. These steps, with the respective counts of observations, are mapped in Appendix A.

Moving from the three data tables to event log, required substantive cleaning and pre-processing to validate, clean, and crosswalk these three datasets. We started this stage with 199,686 **Cases Metadata** rows, 7,631,457 associated docket entries classified by the **Labels Ontology**, and 3,803,194 **Standardized Judges Ontology** rows. After pre-processing, fully outlined in Appendix A, we ended with a complete and validated dataset of 175,268 cases with 4,811,483 timestamped events described by 97 variables. To the best of our knowledge, this is the first event log of U.S. courts and the largest publicly available dataset of validated court operations.

Table 1 presents the structure of the event log, with examples for *Activity attribute* and for a *Case attribute* - Nature of Suit (see complete list of attributes and definitions in Appendix B.)

5. The Federal NDIL Court in Data

In this section we use our constructed event log for descriptive analysis of the key features of the NDIL Court operations. NDIL is among the busiest federal courts in the United States. Each year, it handles thousands of cases, including civil and criminal matters. The court has courthouses in two divisions - the Eastern Division in Chicago and the Western Division in Rockford (indicated by the "city" feature in the dataset).

Figure 5 shows the monthly arrival rate of cases to the NDIL in the years 2004-2018, for civil and criminal cases. We restricted the graph to start in January 2004, as in this year our dataset represents 100% of filed cases and 84.8% of terminated cases reported by the Judicial Business Reports (www.uscourts.gov), and converges to 100% in all of the following years. Pending cases

Figure 5 **NDIL count of cases filed, terminated and pending over time (monthly).**

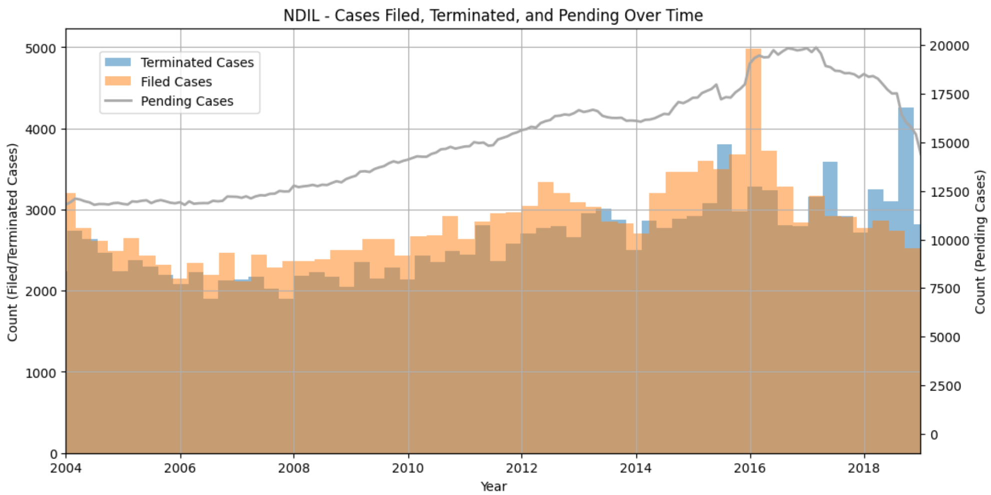

Table 2 **Case attributes distribution in NDIL**

Characteristic	Count	Percentage (%)
Case Type		
Civil	157,703	90.0
Criminal	17,565	10.0
Nature of Suit in civil cases (Top 5 categories)		
Employee Retirement Income Security Act	15,202	8.7
Other Statutory Actions	12,541	7.2
Other Civil Rights	11,974	6.8
Employment	11,553	6.6
Prison Condition	10,346	5.9
Case Status		
Closed	168,107	95.9
Open	7,161	4.1
City		
Chicago	167,687	95.7
Rockford	7,581	4.3

are calculated based on cases that were open during each month shown, as demonstrated by the data.

Similar to the trend for the total system in Figure 1, while there are evident fluctuations in the filed NDIL cases over time, it is clear that the filing rate typically surpasses the termination rate, which explains the accumulation of pending cases over time (also similar to the pattern seen on Figure 1).

As of December 2024, there are 34 active federal district court judges and 14 federal magistrates assigned to the NDIL, which remains fairly stable over the period of our data

Table 3 Party attributes distribution in NDIL

Variable	Mean	Std	Min	Median	Mode	Max
Party counts						
Plaintiffs	1.77	24.41	1.0	1.0	1.0	9,810
Defendants	3.39	22.59	1.0	1.0	1.0	2,755
Plaintiff Counsels	4.55	39.94	0.0	2.0	1.0	14,279
Defendant Counsels	4.17	16.32	0.0	2.0	0.0	4,731
Additional party types (True=1)						
Amicus Curiae	0.00	0.03	0.0	0.0	0.0	1.0
Intervenor	0.01	0.08	0.0	0.0	0.0	1.0
Third Party Defendant	0.01	0.10	0.0	0.0	0.0	1.0
Third Party Plaintiff	0.01	0.09	0.0	0.0	0.0	1.0
Counter Claimant	0.03	0.18	0.0	0.0	0.0	1.0
Counter Defendant	0.03	0.18	0.0	0.0	0.0	1.0
Trustee	0.00	0.04	0.0	0.0	0.0	1.0
Court Monitor	0.00	0.01	0.0	0.0	0.0	1.0
Party attributes (%)						
Share of Individual Plaintiffs	0.51	0.49	0.0	0.5	1.0	1.0
Share of Individual Defendants	0.26	0.39	0.0	0.0	0.0	1.0
Share of Pro Se Plaintiffs	0.13	0.33	0.0	0.0	0.0	1.0
Share of Pro Se Defendants	0.03	0.15	0.0	0.0	0.0	1.0
Share of Pro Hac Vice Plaintiffs	0.03	0.17	0.0	0.0	0.0	1.0
Share of Pro Hac Vice Defendants	0.04	0.18	0.0	0.0	0.0	1.0

Table 4 Descriptive statistics of Length of Stay (in days)

Case Status	Mean	Std. Dev.	Min	25%	Median	75%	Max
Closed	479	641	0	103	267	602	7,599
Open	1,082	1,182	0	575	915	1,158	10,774

(www.ilnd.uscourts.gov). They handle all civil and criminal cases, in various subjects and complexities. Table 2 presents the distribution of the main case attributes in the event log data. In addition, it shows the distribution of the main Nature of Suit categories in civil cases, as well as the case status, where 4.1% of the cases in the dataset were still open in their most recent download (dated mapping of the data pipeline can be found at docs.scales-okn.org). The table also shows the distribution of cases between the two divisions, offering opportunities for comparative studies. Table 3 presents the distribution of the party related attributes. As we explained in subsection 4.3 above, party types are complexity indicators for the individual cases. They range from the average counts of parties and attorneys of different types, to the average share of self represented (pro se) parties, and out-of-district lawyers temporarily admitted to practice in the NDIL (pro hac vice). See Table 7 in Appendix C for definitions of different party types.

Table 5 Summary statistics for each phase of the judicial process in NDIL

	Phase 1	Phase 2	Phase 3
Cases entering the phase	155,525	73,482	2,747
Share of all cases	100%	47.2%	1.8%
Cases completing the phase	74,064	2,747	1,762
Share completed of all cases	47.6%	1.8%	1.1%
Mean activity count	13.84	31.19	53.95
Mean LOS for cases entering the phase	212	554	707
Mean LOS for cases completing the phase	140	706	801
Mean LOS for cases not completing the phase	276	548	541

We next explore the dynamics of the process at the NDIL, as captured by our dataset. First, we present in Table 4 descriptive statistics of the LOS distributions for cases that are closed and open as of their downloading date (see docs.scales-okn.org). For all cases, open or closed, LOS was calculated from the first event to the last event. Several observations are evident from Table 4. First, both distributions are highly positively skewed, with the mean higher than the median, and the maximum values that are extremely large, even compared to the 75th percentile. While the "typical" closed case takes just under 9 months to go through the system (based on the median), the mean is closer to 16 months, with the maximum value of over 21 years. Second, the open cases have much longer LOS than the closed cases (median is over 2.5 years, with the mean of just under 3 years), suggesting the LOS distribution is of 'old better than new' (Englund 2010) type, i.e., older cases are inherently more complex (otherwise they would have been closed).

Next, Table 5 presents summary statistics for the three phases of the judicial process we outline in Figure 3, as they appear in the NDIL dataset. For this analysis, we first define events that mark the shift between phases, based on the activities in subsection 4.3. For the first phase, we set the completion of phase 1 and beginning of phase 2 by either an 'answer' or a 'plea.' We set the completion of phase 2 and beginning of phase 3 by the first 'trial' activity. We set the completion of phase 3 (and the process) by 'verdict' or 'finding of facts'. Note that "LOS" on the last three lines of the table refers to time spent in each phase, measured in days, rather than the total LOS. Excluded from the LOS calculations are cases with censored information due to (i) being still open, (ii) where a phase was inferred by the next phase's event, but lacked a timestamped completion

Table 6 Distribution of disposition types

Disposition category	Case count	Share of total cases
Party resolutions only	73,653	43.8%
Non-trial court decisions only	66,703	39.7%
Party resolution and non-trial court decisions	16,193	9.6%
Trial court decisions only	802	0.5%
Trial court decisions and anything else	1,613	1.0%
Opened in error	77	0.05%
Other	9,066	5.4%

Note: Open cases are not included. 'Other' are either cases with no recorded dispositive event, or where the ML classifiers erroneously missed a dispositive event.

event, (iii) that have LOS=0, or (iv) are transfer cases (either transferred from or to other judicial districts for a variety of reasons).

As noted earlier, the most striking feature of Table 5 is the narrowness of the funnel: only 47.6% of cases make it past phase 1, only 1.8% make it through phase 2, and just 1.1% are resolved by the judge's verdict. This suggests that merely counting the pending cases, without accounting for their distribution between phases, may lead to a misleading representation of the "case load".

Another observation from Table 5 is that progressing through phases does not necessarily imply a longer time spent in a given phase: for cases ending in phase 1 (52.4% of all cases), the average phase 1 LOS is 276 days, which is almost twice the average phase 1 LOS of cases moving forward to phase 2 (140 days). This unique aspect of the process stems from the dynamics of termination in these phases. For example, settlement discussions can delay progress in court because the parties ask for a stay while they negotiate. On the other hand, for cases moving beyond phase 1, both phase 2 and phase 3 LOS for cases completing the relevant phase is larger than for those not completing it - this may point to congestion-related delays of waiting for the trial date.

The "vanishing trial" phenomenon ((Galanter 2004), where only a very small share of cases are resolved at the trial phase is also evident from Table 6, which shows the distribution of cases among the different disposition types. To capture the diverse ways in which cases are resolved, we defined the disposition categories based on the entire life cycle of a case, recognizing that different components, such as parties or claims, can be terminated from cases in distinct ways. This classification provides valuable insight into the disposition process. It addresses the timing of case disposition and the mechanisms through which cases are terminated. Understanding these dynamics complements the case funnel by highlighting the multifaceted nature of case outcomes and the decomposition of the case throughout its life cycle.

Finally, section D presents the data-driven process map, created with the *Disco* process mining tool (Günther and Rozinat 2012). The map demonstrates both the complexity of the underlying process and the potential for process-mining and data-driven simulation tools to derive insights from our data set.

6. Research Directions

As discussed in Section 2, despite growing interest in applying OR/MS methodologies to judicial systems, empirical research in this field has largely focused on simplified judicial environments with well-defined constraints and relatively low procedural complexity. Prior studies address court congestion and resource allocation (e.g., (Bakshi et al. 2024, Gupta and Bolia 2023, Peyrache and Zago 2024)) and explore potential interventions, such as dedicated dockets (Freund and Weng 2024). However, these efforts tend to overlook key factors driving court operations, such as the strategic behavior of judges and parties. This gap is due in part to scarcity of granular data, critical for capturing the nuanced dynamics of judicial operations (Dimas et al. 2023, Pah et al. 2020).

The current paper constructs a dataset that enables a deeper exploration of court efficiency by examining, for example, Length of Stay (LOS) as a primary performance measure (Marciano et al. 2019). We propose three levels of analysis to investigate key research questions: case-level, judge-level, and system-level.

At the case level, the overriding question is the extent to which various case characteristics influence how quickly the case moves through the system and what interventions are likely to be effective in facilitating lower LOS. We hypothesize that case complexity plays a central role in shaping court operations. Prior literature highlights case attributes as strong predictors of LOS (de Oliveira et al. 2022, Gruginskie and Vaccaro 2018, Vercosa et al. 2023). Some delays may stem from inherent case characteristics such as the number of involved parties, claim complexity, and evidentiary requirements, while others arise due to procedural inefficiencies or intentional delay tactics. This dataset provides an opportunity to analyze the impact of these factors on LOS and case resolution pathways. Additionally, it allows for differentiating between exogenous (structural) and endogenous (strategic) delays across various case stages. Moreover, the impact of procedural interventions, such as case-level FIFO scheduling (Azaria et al. 2023, 2024, Bray et al. 2016), can be more accurately modeled to estimate their effectiveness in expediting case resolution.

At the judge level, the main research questions are quantifying the extent to which judicial decision making and case management strategies influence various outcomes, ranging from the LOS of cases of different types, settlement rates, and final rulings. Previous research (Azaria and Shamir 2024, Coviello et al. 2015, Engel and Weinshall 2020, Kricheli-Katz and Weinshall 2023) indicates that reduced workload and delay improves the court performance in terms of justice, operational quality and judicial efficiency. The granularity of our data should enable answers to several questions, including: What are the impacts of the workload level and the workload composition a judge faces on the outcomes listed above? In addition, a judge's experience with specific case types, similar to skill-based routing, may contribute to variations in LOS (Castelliano et al. 2021). We note that traditional measures of judicial efficiency, such as clearance rates, often fail to capture the complexities of judicial workload. This dataset offers an opportunity to assess efficiency metrics that account for rework, prolonged decision processes, and workload dynamics over time. Tracking how judges interact with different case types, identifying behavioral patterns, and assessing decision-making trends can provide deeper insights into the impact of specialization. Observing procedural variations across cases handled by different judges may reveal underlying efficiencies or inefficiencies, allowing for a more nuanced understanding of judicial performance.

At the system level, broader systemic factors likely contribute to court congestion and inefficiencies. Understanding LOS trends is crucial for identifying conditions under which the system becomes overloaded. Prior research suggests that case assignment mechanisms influence court efficiency and fairness. Empirical studies on case allocation in various judicial settings indicate that specialized judges, who focus on specific case types, tend to process cases more efficiently than those assigned randomly (Macfarlane 2024, Marcondes et al. 2019). However, specialization may also introduce biases or inefficiencies if not managed properly.

Beyond judge assignment, procedural changes influence court workload and overall demand. Modifications such as alternative dispute resolution mechanisms can significantly impact the number and type of cases entering the system. These procedural shifts affect not only LOS but also resource allocation and congestion levels. Insights from Load Effect on Service Time (LEST, (Delasay et al. 2019)) literature suggest that congestion significantly affects LOS in service environments, and applying these principles to judicial systems could provide new perspectives on managing caseloads effectively.

By leveraging a detailed event log, this dataset provides an opportunity to rigorously analyze these questions across multiple dimensions. Future research can apply queue mining and simulation techniques to uncover causal relationships and evaluate potential interventions. These insights could inform data-driven policies to enhance efficiency, reduce backlogs, and improve access to justice.

7. Conclusions

We present a novel and comprehensive dataset derived from 18 years of federal district court case dockets, with detailed event logs designed for operations research applications. We provide tools for analyzing judicial operations and addressing critical challenges such as congestion, resource allocation, and inefficiencies in case flow.

The dataset's structure, integrating metadata, timestamps, and activities, enables researchers to explore the complexities of service systems characterized by high variability and long durations.

This work offers significant contributions to the operations research and law and policy communities, fostering data-driven approaches to improve access to justice and optimize court performance. The dataset not only supports theoretical exploration but also serves as a practical foundation for designing interventions with tangible social impact.

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Appendix A: Data Funnel

Creating the event log is a considerable challenge (van der Aalst et al. 2012). Figure 6 shows various stages of the event log construction process, and the exclusion decisions that were taken along the way, illustrating the complexity of dealing with this challenge. The graph demonstrates the path of the datasets provided by subsection 4.2, leading to the unified event log dataset presented in this paper. This path could be used as a general scheme for cleaning, validating, and merging similar data sources.

Appendix B: Activities and Attributes Definitions

The list of 26 Activities below was carefully constructed in collaboration with domain experts, but with a process view in mind. While activities are a higher level description of events in the case life cycle, attributes are paired with activities and provide additional detail pertinent to that specific occurrence. Some attributes pertain only to certain specific activities, others are "floating" attributes and can be attached to any activity or larger categories of activities.

Below is the list and definitions of the activities in our event log. Some activities are relevant only for one type of cases (e.g. civil, criminal), as indicated in their definition. For activities with complementing attributes (i.e. specific types of an activity), a sub-list of attributes and definitions is provided.

1. *Complaint*: In civil cases, an initial pleading filed by a plaintiff that outlines the facts of the case and legal claims against a defendant. In criminal cases, a document that establishes probable cause that a crime was committed and the defendant is likely responsible.

2. *Removal*: The removal of a case initially filed in state court to federal court.

3. *Petition*: A formal written request submitted to a court, asking for a specific action or order, often used to initiate certain types of legal proceedings.

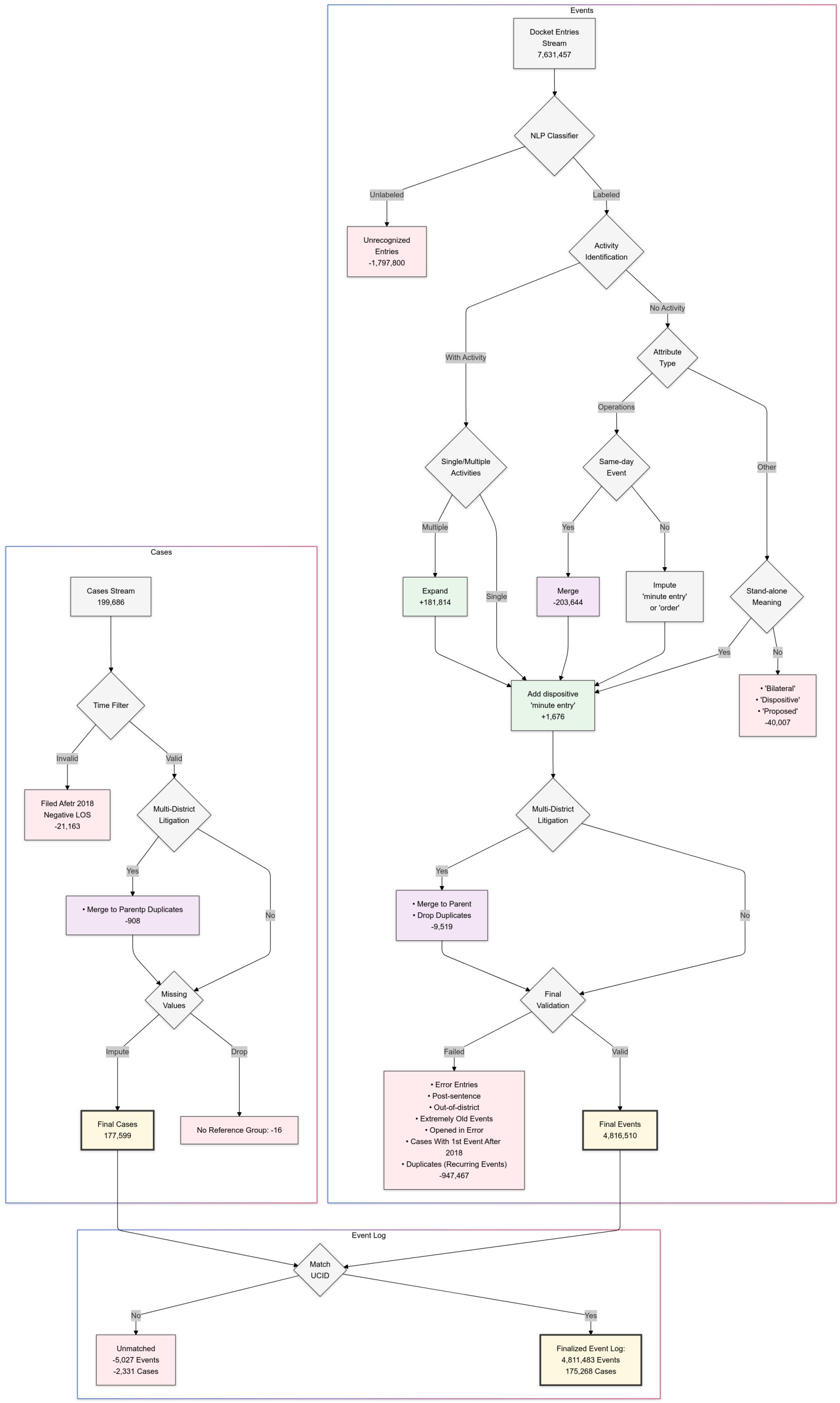
- *Petition for Habeas Corpus*: A legal action challenging the legality of a person's detention or imprisonment, seeking a court order for their release if the detention is deemed unlawful.

4. *Summons*: A legal document issued by a court, informing a person that they are required to appear in court or respond to a legal action

5. *Warrant*: A legal document issued by a judge or other judicial authority, authorizing law enforcement to perform specific actions, such as conducting a search or making an arrest.

6. *Arrest*: Documentation of arrest.

Figure 6 Data funnel.



7. *Information*: A formal charge, similar to an indictment, but filed by a prosecutor instead of a grand jury, initiating a criminal case.

8. *Indictment*: A formal charge issued by a grand jury, accusing a defendant of committing a crime, initiating a criminal prosecution.

9. *Plea*: A criminal defendant's response to the charges against them.

- *Plea Guilty*: A criminal defendant's response of guilty to the charges against them.

- *Plea Not Guilty*: A criminal defendant's response of not guilty to the charges against them.

10. *Waiver*: A party's voluntary relinquishment of a legal right, claim, or privilege.

- *Waiver of Indictment*: A voluntary agreement by a defendant to waive their right to have a grand jury review the criminal charges against them.

11. *Answer*: A defendant's formal response to a plaintiff's claims, addressing and contesting the allegations in a complaint.

12. *Motion*: A formal request submitted to the court by a party in a legal proceeding, asking for a specific action, ruling, or order.

- *Motion for Arbitration*: A motion seeking dismissal of the case for purposes of arbitration.

- *Motion for Default Judgment*: A motion made by a plaintiff for a judgment in their favor due to the defendant's failure to respond or appear in court.

- *Motion for Habeas Corpus*: A legal action challenging the legality of a person's detention or imprisonment, seeking a court order for their release if the detention is deemed unlawful. These docket entries are most likely the same as petitions for habeas corpus but named differently by the filing party (a prisoner seeking release). Many prisoners are not represented by lawyers (pro se) and therefore use varying naming conventions for their filings.

- *Motion for Judgment as a the Matter of Law*: A motion made during a trial for the court to decide in their favor based on the evidence presented, arguing that no reasonable jury could reach a different conclusion.

- *Motion for Judgment on the Pleading*: A motion for a court to decide a case based solely on the pleadings filed.

- *Motion for Summary Judgment*: A motion for the court to decide a case without a trial when there are no genuine disputes of material fact.

- *Motion for Judgment (other)*: Motion for judgment in a case in full or in part that does not fall into any of the other judgment-related motion types, e.g., motion for summary judgment or judgment as a matter of law.

- *Motion for Settlement*: A motion for the court to take some action, often dismissing the case in full or in part, as a result of the parties' settlement.
- *Motion for Time Extension*: A motion asking the court to extend a deadline or time limit.
- *Motion for Certify Class*: A motion to certify the case as a class action.
- *Motion to Remand*: A motion asking the court to send a case to a lower court or administrative body for further action or reconsideration.
- *Motion to Seal*: A motion requesting the court to protect sensitive or confidential information in a case by restricting public access to specific documents or portions of the court record.
- *Motion to Dismiss*: A motion requesting the court to terminate a case in full or in part. (The vast majority of motions receiving this label are contested motions filed under FRCP 12(b) or 41(b). However, some docket entries that receive this label may in fact be voluntary motions filed under FRCP 41(a) or filed as a result of a settlement, but that information is not present on the face of the docket sheet.) When a docket entry contains only bare "motion to dismiss" language, with no other indication of which type of motion it is, the motion to dismiss attribute label is applied as a default. When some indication of voluntariness is present in the docket entry, the relevant voluntary dismissal label is applied instead.
- *Motion for Voluntary Dismissal*: A document filed by a party to voluntarily dismiss their case or specific claims or counter-claims. Voluntary dismissal itself may or may not mean that a settlement has occurred. Where the docket entry also contains signals of settlement, the separate settlement labels are applied. Where no separate indications of settlement appear on the docket sheet, researchers may choose how to treat "bare" voluntary dismissals vis-a-vis settlement.
- *Motion for Dismissal (other)*: Motion for dismissal of a case in full or in part that does not fall into any of the other dismissal-related motion types, e.g. motion for voluntary dismissal or contested/adversarial motions to dismiss.

13. *Brief*: A document that accompanies a motion and contains support for the motion.
14. *Notice*: A notification by the parties or court.

- *Notice of Appeal*: A formal notification filed by a party to inform the court and other parties of their intention to appeal a decision or judgment to a higher court.
- *Notice of Consent*: A document filed by a party to indicate consent to some action in the case.

- *Notice of Motion*: A document filed by a party that notifies the court and other parties of the intention to seek a specific court order or ruling, usually accompanied by the motion and supporting documents.
- *Notice of Settlement*: A document filed by a party to inform the court of a settlement.
- *Notice of Voluntary Dismissal*: A document filed by a party to voluntarily dismiss their case or specific claims or counter-claims. In civil cases, voluntary dismissal itself may or may not mean that a settlement has occurred. Where the docket entry also contains signals of settlement, separate settlement labels are applied.
- *Notice of Dismissal (other)*: Notice of dismissal of a case in full or in part that does not fall into any of the other dismissal-related notice types, e.g., notice of voluntary dismissal.

15. *Response*: A party's written opposition, response, or reply to a document or motion filed by another party.

16. *Stipulation*: A document that indicates an agreement between parties that establishes facts or terms to be accepted as true or binding.

- *Stipulation of Dismissal*: An agreement between the parties to dismiss a case in full or in part.
- *Stipulation of Judgment*: An agreement between the parties for entry of judgment in a case in full or in part.
- *Stipulation of Settlement*: An agreement between the parties to resolve a case by settlement in full or in part.

17. *Order*: A court decision on a matter in a case or instructions/mandate to the parties.

18. *Minute Entry*: A notation or other communication on the docket sheet recorded by the judge or court clerk.

The below attributes could be attached to either *Order* or *Minute Entry* events:

- *Transfer Inbound*: The transfer of a case from one court to another. The "inbound" label applies to entries indicating a transfer on the docket sheet of the receiving court.
- *Transfer Outbound*: The transfer of a case from one court to another. The "outbound" label applies to entries indicating a transfer on the docket sheet of the sending court.
- *Transfer Unknown*: The transfer of a case from one court to another. The "unknown" label applies to entries where it is unclear from the entry itself whether the present court is the sending or receiving court.
- *Administrative Closing*: A procedural action taken by the court to remove a case from its active docket, often but not always pending finalization of the parties' settlement discussions.

- *Default Judgment*: A decision granting default judgment to a plaintiff.
- *Granting Motion for Summary Judgment*: A court order granting a party's motion for summary judgment in full or in part.
- *Voluntary Dismissal*: An action that dismisses a case pursuant to a party request for voluntary dismissal.
- *Dismissal (other)*: Actions that dismiss a case in full or in part but do not fall into any of the other dispositive event categories, e.g., sua sponte dismissals by the court.
- *Granting Motion to Dismiss*: A court order granting a party's motion to dismiss in full or in part.
- *Remand*: A court's decision to send a case to a lower court or administrative body for further action or reconsideration.

19. *R&R*: A report and recommendation issued by a magistrate judge to a district court judge after consideration of a pending motion or other matter in the case, offering a non-binding recommendation for how the district court judge should rule.

20. *Judgment*: The document that formalizes the final adjudication or resolution of a case.

21. *Settlement*: An indication on the docket sheet that the parties have resolved the case by settlement, in full or in part.

- *Settlement Consent Decree*: A court-approved agreement between parties to resolve a case. Includes docket entries labeled with variations on the term "Consent Decree," including e.g. Consent Judgment.
- *Settlement Rule 68*: An accepted offer of judgment under FRCP 68, which allows a defendant to make a formal settlement offer to the plaintiff, potentially shifting the costs of litigation if the plaintiff ultimately recovers less than the offer amount.

22. *Plea Agreement*: A negotiated agreement between a defendant and a prosecutor in a criminal case in which a defendant pleads guilty or no contest to a criminal charge in exchange for concessions from the prosecutor.

23. *Trial*: A formal legal proceeding where parties present evidence and arguments to a judge or jury to determine the outcome of a case, either in a criminal prosecution or a civil lawsuit. The "trial" label attaches to any docket entry that indicates that trial has begun in the case.

- *Bench Trial*: A trial where the case is decided by a judge rather than a jury.
- *Jury Trial*: A trial where the case is decided by a jury rather than a judge.

- *Trial (other)*: A trial where the face of the docket sheet is unclear whether the trial is a jury or bench trial.

24. *Finding of Facts*: Short for "Findings of Fact and Conclusions of Law." A court's written determination of the facts and conclusions of law relevant to a case, usually based on evidence presented during a bench trial or hearing.

25. *Verdict*: The final decision made by a judge or jury at the close of a trial.

26. *Sentence*: The final judgment of punishment imposed by a court in a criminal case.

Below is a list of the floating attributes, which can be assigned to any activity (unless mentioned otherwise). These attributes are generally related to the routing of the case, indicating the role of the activity rather than its type:

- *Opening*: The event that opens the case in federal court. Could be attached only to specific events (complaint, removal, petition, and some types of order and minute entry).

- *Dispositive*: An action by the parties or the court that ends a case in full or in part. Because these include partial resolutions, multiple dispositive events can be identified in a single docket sheet. Could be attached only to specific events (settlement, finding of facts, verdict, and some types of order and minute entry). Judgments are not considered dispositive in the labeling scheme, as they formalize a separate, dispositive event such as a granted motion for summary judgment. Therefore, the "dispositive" attribute attaches to the separate dispositive event rather than to the "judgment" event. Likewise, sentences in criminal cases are not labeled as dispositive, as they are the follow-on event after a defendant has been found or plead guilty, the act that resolved the pending charges.

- *Scheduling*: An action by the court that sets a deadline or a date for future activity in the case.

- *Rescheduling*: An action by the court that changes an existing deadline or date for future activity in the case.

- *Hearing Conference*: A hearing, conference, meeting, or other communication between the judge and the parties.

- *Bilateral Unopposed*: Docket entry language that indicates agreement between/among the parties to the action described in the docket entry.

- *Proposed*: A document or action filed by the parties in "proposed" form for the court to consider.

- *Case Opened in Error*: Notice by the court that the case was opened in error.

- *Transferred Entry*: Docket entries that were copied from one court's docket sheet to another to capture activity pre- and/or post-transfer between courts.
- *Error*: Notice by the court of an error in filing that docket entry.
- *Dismiss With Prejudice*: Docket entry language indicating that a dismissal is with prejudice, foreclosing further action in the case. Could be attached to and dismissal- or settlement-related events.
- *Dismiss Without Prejudice*: Docket entry language indicating that a dismissal is without prejudice, allowing further action in the case. Could be attached to and dismissal- or settlement-related events.

Appendix C: Variables Description

Table 7 presents the variables in the data set, their type, description and a sample value. The rest of the variables in the data are Boolean variables that indicate specific attributes related to each event.

Appendix D: Process Map

Figure 7 presents the data-driven process map, created with the *Disco* process mining tool (Günther and Rozinat 2012). This map illustrates the flow of a sample of civil cases at the Rockford Division of the NDIL. While it is filtered to show only the most common paths for simplicity, it demonstrates the process flow according to the data. The map tracks the flow of cases between activities, starting with the green arrow at the top and ending with the red square at the bottom. Shades of the activities represent their frequency in the data (darker are more frequent) as do the thickness or the arrows for flow (thicker arrows for more cases follow this path). It should be noted that due to the filtering of the paths, the count of cases presented by the numbers in the chart is partial.

Table 7 **Variables and their definitions** **(continued on next page)**

Field Name	Data Type	Description	Sample Value
ucid	String	Unique case ID, pending the court abbreviation with Pacer case ID	ilnd;;1:00-cr-00238
date_filed	Date	The date of the docket entry	7/16/2016
event_judge	List	Judges SJID and type, recognized in the event text	('SJ000152', 'District Judge')
Activity	String	Type of legal or procedural actions recorded in the docket entry	motion
attribute_duplicates	integer	Count of duplicate events of the same type (returns)	1
city	string	One of the two NDIL divisions - Chicago or Rockford	Chicago
case_type	string	General type of proceedings (either criminal or civil)	cv
case_status	string	Indicates weather the case is 'closed' in the data or 'open'	closed
nature_suit	string	A code indicating the category of the procedure (civil only)	550 Civil Rights
is_mdl	boolean	Indicates whether this case is part of a multi-district litigation	True
plaintiffs_count	integer	Total number of plaintiffs per case	1
plaintiffs_share_ind	float	Share of plaintiffs that are individuals	1
plaintiffs_share_pro_se	float	Share of self-represented plaintiffs	1
plaintiffs_share_pro_hac_vice	float	Share of plaintiffs with out-of-district counsels	1
plaintiffs_counsels_count	integer	Total number of counsels for the plaintiffs	1
Defendants_count	integer	Total number of defendants per case	1
Defendants_share_ind	float	Share of defendants that are individuals	1
Defendants_share_pro_se	float	Share of self-represented defendants	1
Defendants_share_pro_hac_vice	float	Share of defendants with out-of-district counsels	0.67
Defendants_counsels_count	integer	Total number of counsels for the defendants	1
Defendants_highest_offense_opening	string	Highest level offense filed against the defendant	Felony
Defendants_highest_offense_terminated	string	Highest level offense of the defendant which was terminated	Misdemeanor
Defendants_pending_counts	integer	Number of pending charges against a defendant	1
Defendants_terminated_counts	integer	Number of terminated charges against a defendant	1
Party_Amicus	binary	Indicates presence of amicus curiae	1,0
Party_Counter_Claimant	binary	Indicates presence of party filing counterclaim	1,0
Party_Counter_Defendant	binary	Indicates presence of counterclaim defendant	1,0
Party_Court_Monitor	binary	Indicates presence of court-appointed monitor	1,0
Party_Intervenor	binary	Indicates presence of intervenor	1,0
Party_Material_Witness	binary	Indicates presence of material witness	1,0
Party_Third_Party_Defendant	binary	Indicates presence of third-party defendant	1,0
Party_Third_Party_Plaintiff	binary	Indicates presence of third-party plaintiff	1,0
Party_Trustee	binary	Indicates presence of trustee	1,0
Other_courts	binary	Indicates involvement of other courts	1,0
related_case_count	integer	Number of cases defined as related	10

Figure 7 NDIL data-driven process map (Civil cases, Rockford Division, 2012-2013).